

DragonWings Power Availability Profile Forecaster

A bounded agent for additional solar power estimates

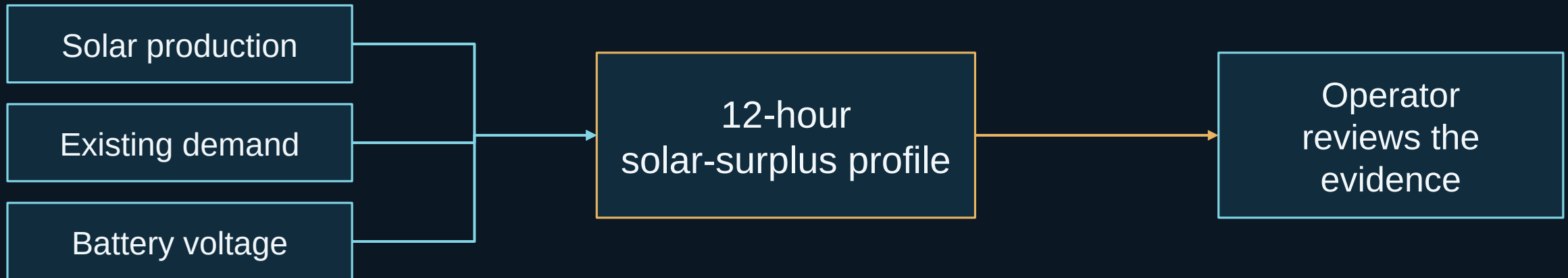
Working MVP with inspectable evidence and human oversight

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The operator's planning question

How much extra load might the next 12 hours support?



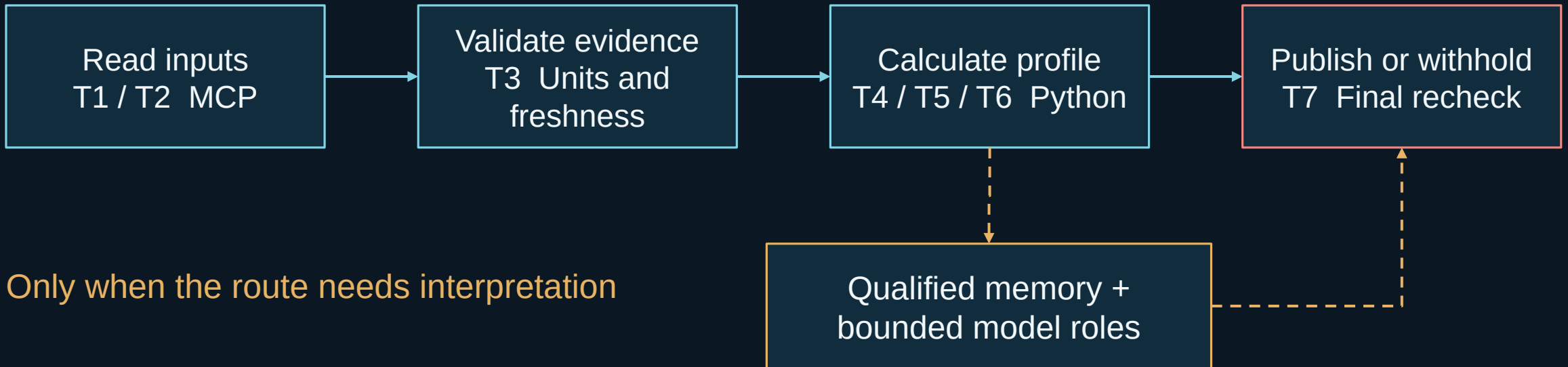
What it does

Estimates additional power and energy by interval.

Why it matters

A reading taken now does not establish future availability.

One workflow connects the components



Only when the route needs interpretation

PostgreSQL stores evidence and results. LangGraph checkpoints store workflow progress.

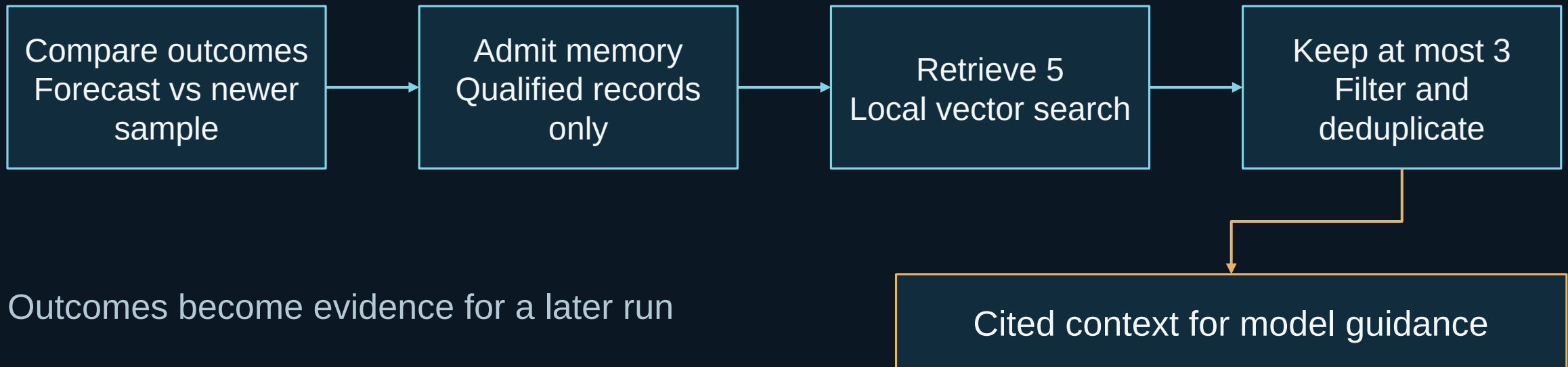
What it does

Runs the evidence-to-decision sequence automatically after a request.

Why it matters

Every result can be traced to inputs, calculations, and a stopping decision.

Memory adds relevant experience



Outcomes become evidence for a later run

What it does

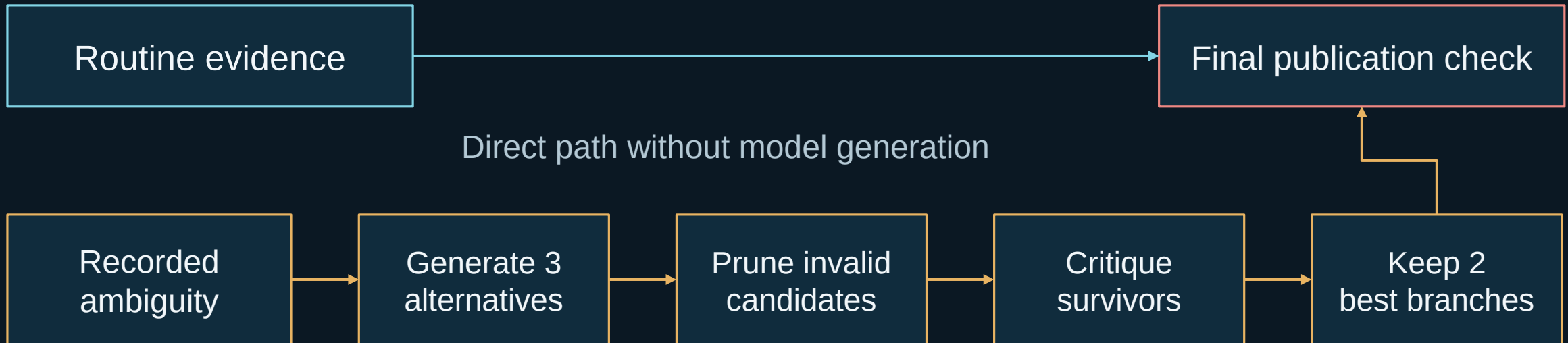
Finds relevant past outcomes and operating guidance.

Current measurements and numerical constraints stay authoritative

Why it matters

The agent can explain caution using evidence from an earlier run.

Extra reasoning has a fixed budget



One refinement. Depth-three submission. Eight shared model attempts maximum.

What it does

Compares a few explanations when recorded evidence justifies it.

Why it matters

Simple cases stay quick, and difficult cases still terminate.

The third role added cost in one trial



Same evidence snapshot	Model calls	Elapsed	Result
Two roles	4	53.035 s	Valid
Three roles	6	70.223 s	Withheld

Calculated kW and kWh stayed identical

What it does

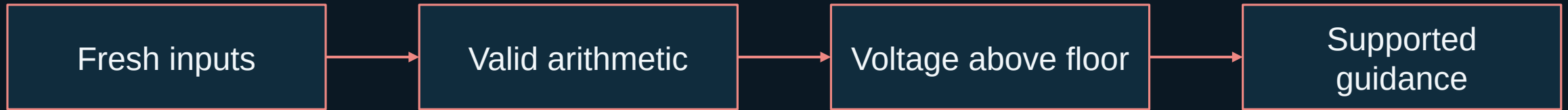
An environment flag enables an additional interpretation role.

Why it matters

Its contribution can be compared before accepting added complexity.

One sequential pair. No demonstrated quality improvement. Third role defaults off.

Validation controls publication



305.2 V

Fixed observed voltage floor

Pass: publish read-only profile

Fail: withhold with a reason

What it does

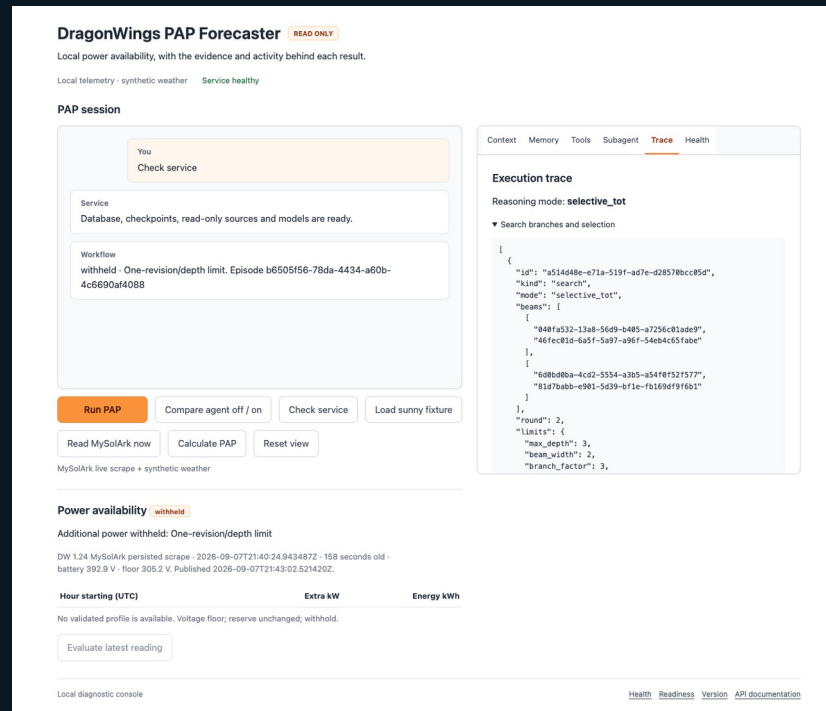
Stops publication when required evidence or checks fail.

No equipment commands. No prediction or guarantee of future battery voltage.

Why it matters

A convincing explanation must not bypass an operating constraint.

Six views explain each decision



Context

Inputs and calculated facts

Memory

Retrieved records and feedback

Tools

Schemas, calls, and results

Subagent

Role inputs and validated outputs

Trace

Branches, scores, and stop reasons

Health

Source and dependency readiness

What it does

Makes the working agent inspectable from one local console.

Why it matters

A result is easier to debug when its evidence and decisions are visible.

Evidence supports the integrated workflow

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Unit and integration tests

Reproducible local gate

What it does

Tests arithmetic, boundaries, persistence, retrieval, and the browser flow.

Required tests use model doubles. Live checks are small demonstrations, not accuracy rates.

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Chromium
Playwright checks

One newer real telemetry sample

Point-power metric	Error
Solar overestimation	0.410 kW
Availability error	0.076 kW

Why it matters

Separates software correctness from forecast and advice quality.

Repository and next steps

<https://github.com/roboyak/PAP-Agent>

README.md

Setup and usage

src/pap_agent/

Workflow, tools, models, and UI

tests/ and docs/pr/

Verification and teaching evidence

submission/

Editable content and artifact builders

make bootstrap

PAP_PROFILE=macmini-replay make run

make verify

What it does

A reproducible local MVP with bounded, observable model roles.

Next improvements

Validated weather and equipment limits

Held-out forecast outcomes

Repeated agent comparisons and operator review

Why it matters

The next improvements can be measured against an inspectable baseline.